Uber open sources Web based platform for vehicle data

Uber has announced that it is open sourcing its autonomous visualisation system (AVS), which it describes as a new way to understand and share autonomous systems data.

"AVS is a new standard for describing and visualising autonomous vehicle perception, motion, and planning data, offering a powerful Web based toolkit to build applications for exploring, interacting and, most critically, making important development decisions with that data," an Uber blog post stated.

"As a standalone, standardised visualisation layer, AVS frees developers from having to build custom visualisation software for their autonomous vehicles. With AVS abstracting visualisation, developers can focus on core autonomy capabilities for drive systems, remote assistance, mapping and simulation," the blog post added.

The advanced technologies group (ATG) and visualisation teams at Uber have already leveraged AVS to support an ever-growing pool of autonomous use cases, such as root cause analysis, Web based log-viewing, developer environments and mapping maintenance.

Now, several other companies, including Voyage and Applied Intuition, have also pledged to use it.

The company says AVS' architecture distinguishes itself from current solutions in many ways.

- It has been designed with an intentional separation of data from any underlying platform.
- Its limited specifications make tools easier to develop.
- Its data format requirements result in fast transfer and processing

AVS was created to meet the needs of everyone in the autonomous visualisation ecosystem, including engineers, vehicle operators, analysts, and specialist developers.

"Autonomous engineers can describe their systems with XVIZ easily, and then test and visualise their expectations with limited overhead. Specialist developers can quickly build data source-agnostic applications with strong performance characteristics and simplified integration using Streetscape.gl. Lastly, operators can view the data in standard visual formats, including videos, across multiple applications, leading to easier collaboration, knowledge-gathering, understanding, deeper analysis, and overall trust in the data quality," a company spokesperson explained.

NSA open sources Ghidra – a software reverse engineering tool

The United States' National Security Agency (NSA) has announced the release of Ghidra, a software reverse engineering (SRE) framework developed by the agency's Research Directorate for its cyber security mission, under the Apache 2.0 licence.

The tool helps analyse malicious code and malware like viruses, and enables cyber security professionals to gain a better understanding of potential vulnerabilities in their networks and systems.

Rob Joyce, the cyber security adviser to the NSA director, made the announcement at the 2019 RSA security conference, and called it a "contribution to the nation's cyber security community."

Joyce said that the tool was internally developed to take a deep look into malware and software to spot the weak points and exploit them. It also lets multiple users reverse engineer the same binary at the same time.

Its capabilities include disassembly, assembly, decompilation, graphing and scripting. It supports a wide variety of processor instruction sets and executable formats, and can be run in both user-interactive and automated modes.

The tool can run on a variety of platforms including Windows, MacOS and Linux. The users can also develop their own Ghidra plugin scripts using the available API.

At the moment, it can be downloaded from the official website of Ghidra. NSA also plans to release its source code under an open source licence on GitHub.